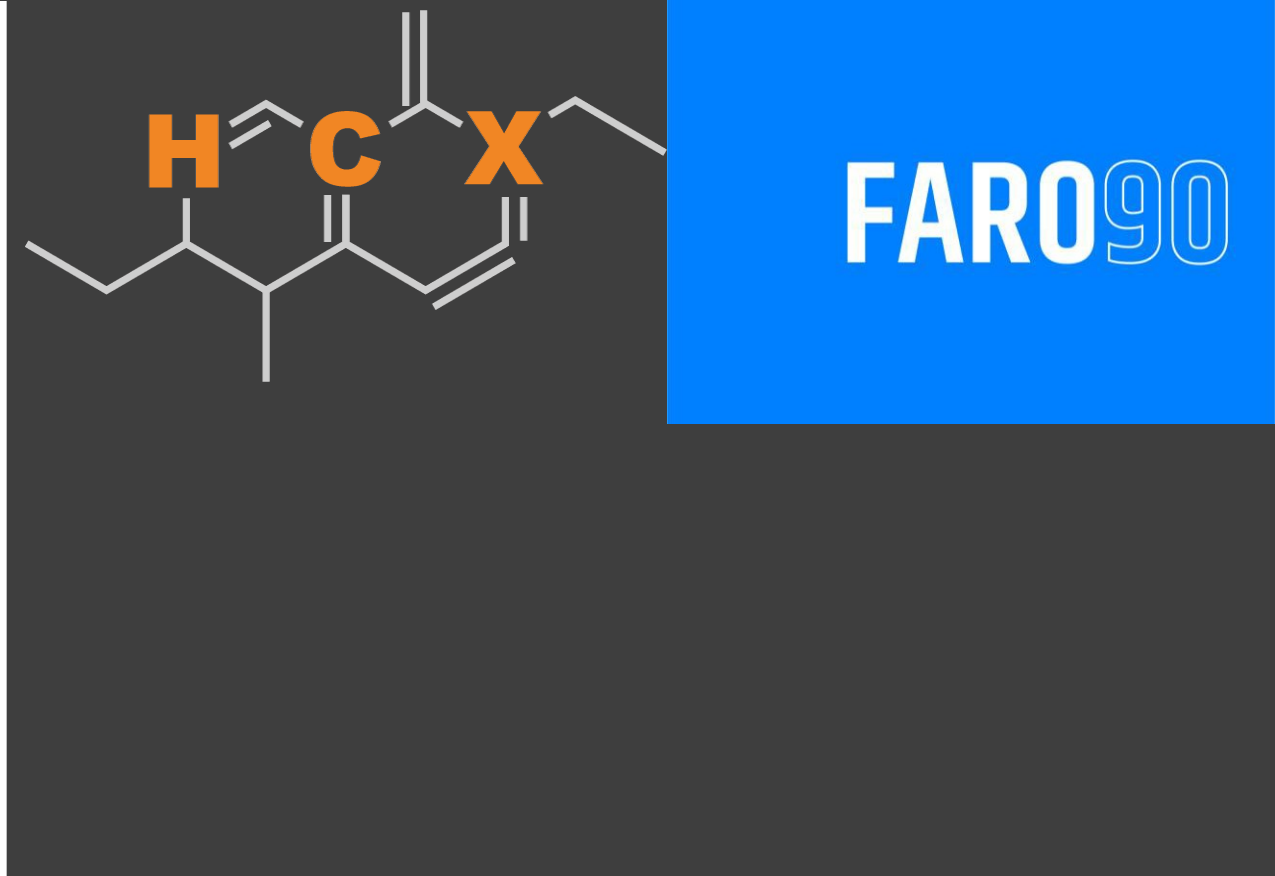




Ethanol Blending in Gasoline - Finland

September, 2025

Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Finland



In 2024, gasoline consumption in Finland was 2,341 million liters (618 million gallons) and growth rate was 0.5%. Gasoline production and net imports were 5,450 and -3,109 million liters (1,440 and -821 million gallons) correspondingly. Finland complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

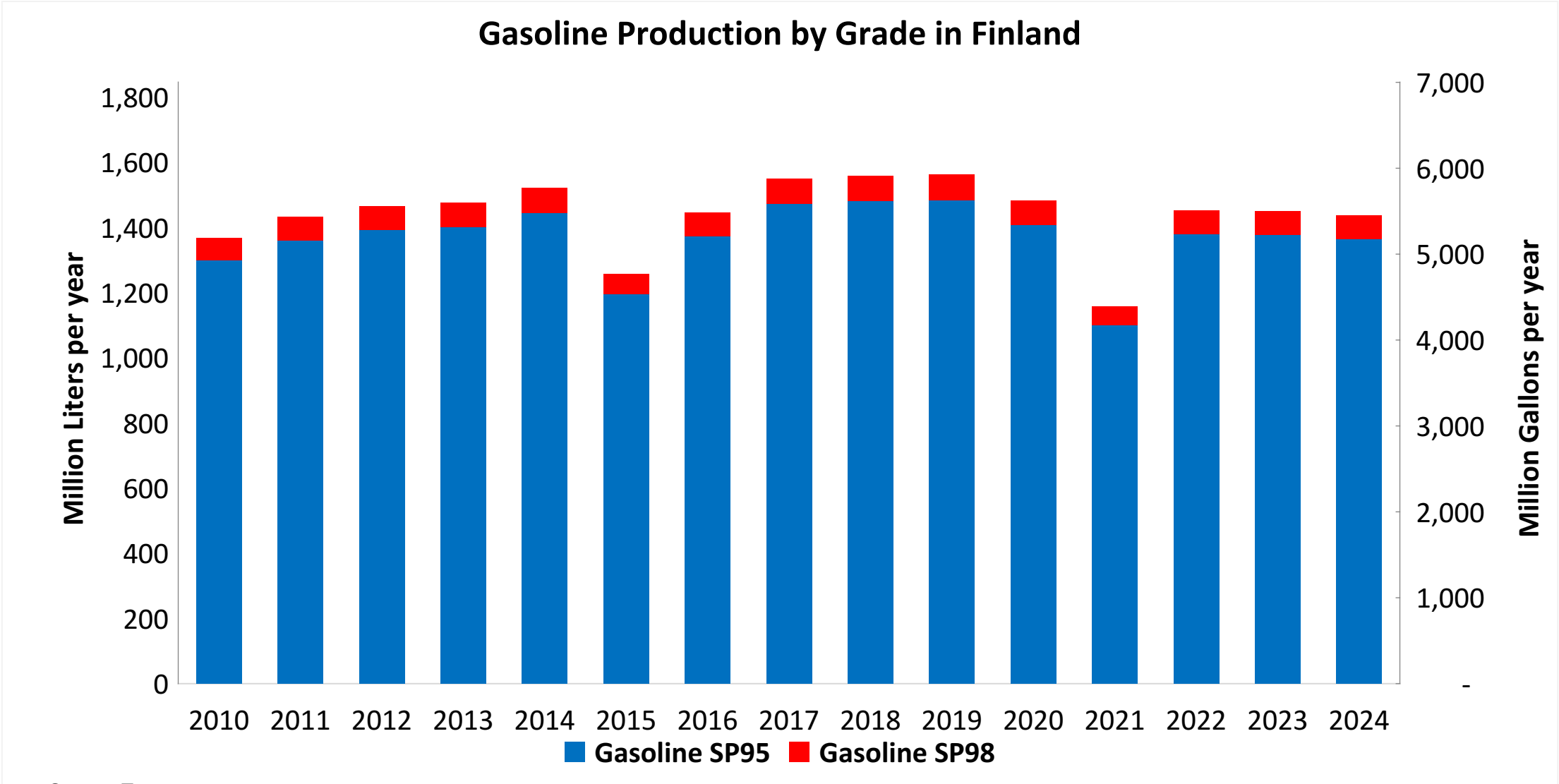
In 2024, ethanol consumption in Finland was 299 million liters (79 million gallons) representing 12.8% of gasoline demand. Ethanol production and Net Imports were 67 and 233 million liters (79 and 61 million gallons) correspondingly. Ethanol sources are Corn 5%, Cereals 85% and 2G lignocelulosic Ethanol 10%.

Source: Eurostat, European Commission

The FARO90 logo is shown in a blue box, followed by a chemical structure diagram of a substituted cyclohexene. The structure features a six-membered ring with a double bond. Substituents include a methyl group, a propyl group, and a group labeled 'X'. The letters 'H', 'C', and 'X' are highlighted in orange in the original image.



Gasoline Production in Finland

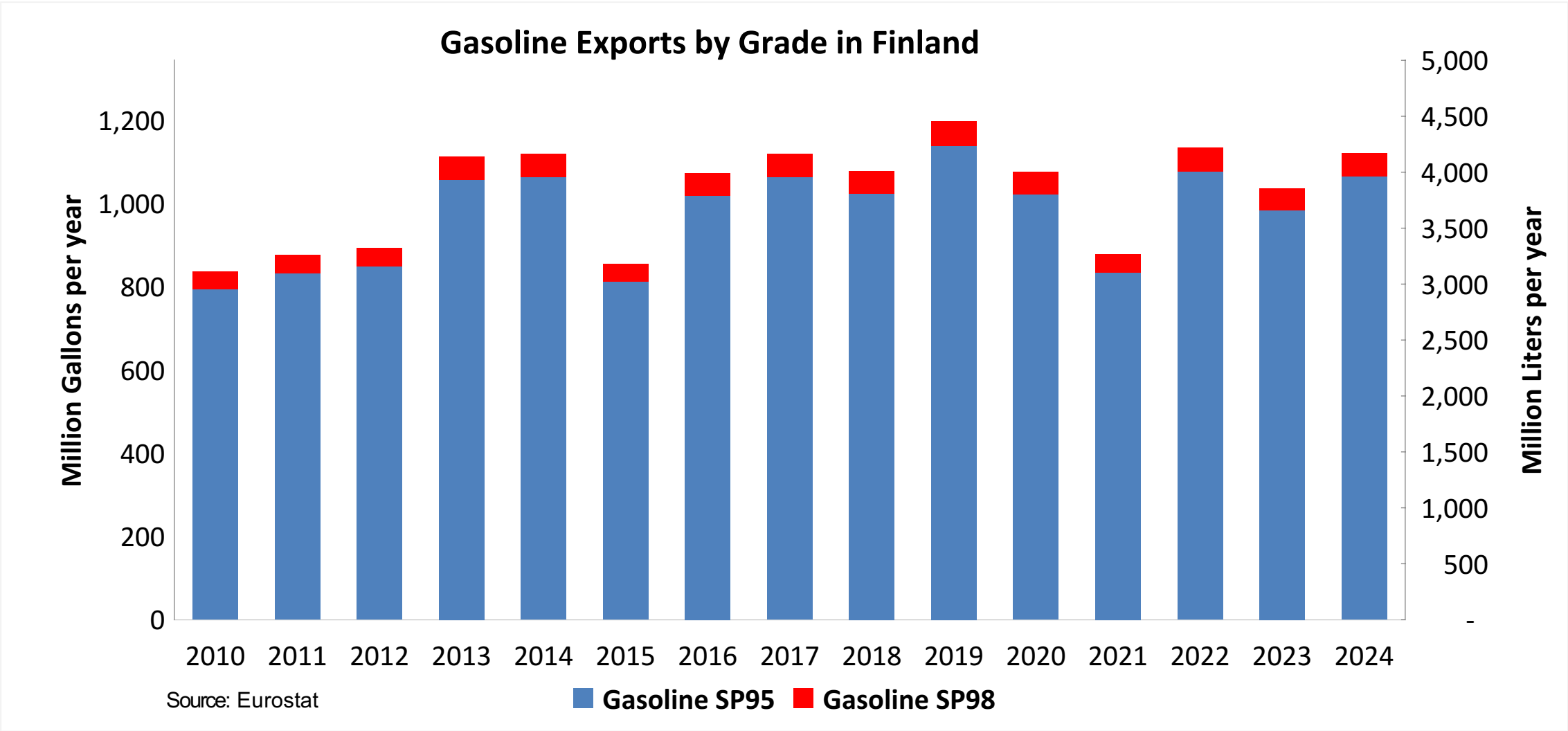


Source: Eurostat

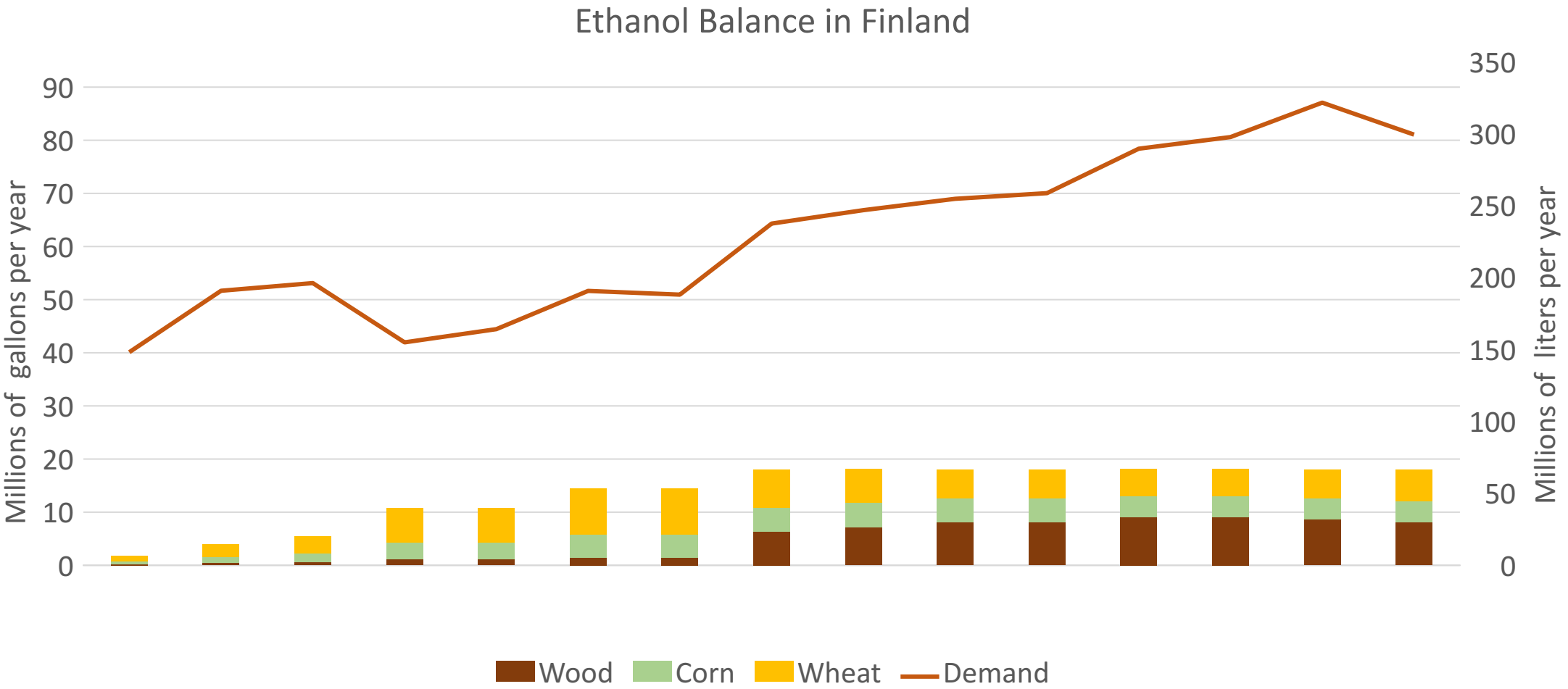
The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a methyl group, a propyl group, and a group labeled 'X'. The letters 'H', 'C', and 'X' are highlighted in orange in the original image.



Gasoline Exports in Finland



Ethanol Balance in Finland



Source: Eurostat

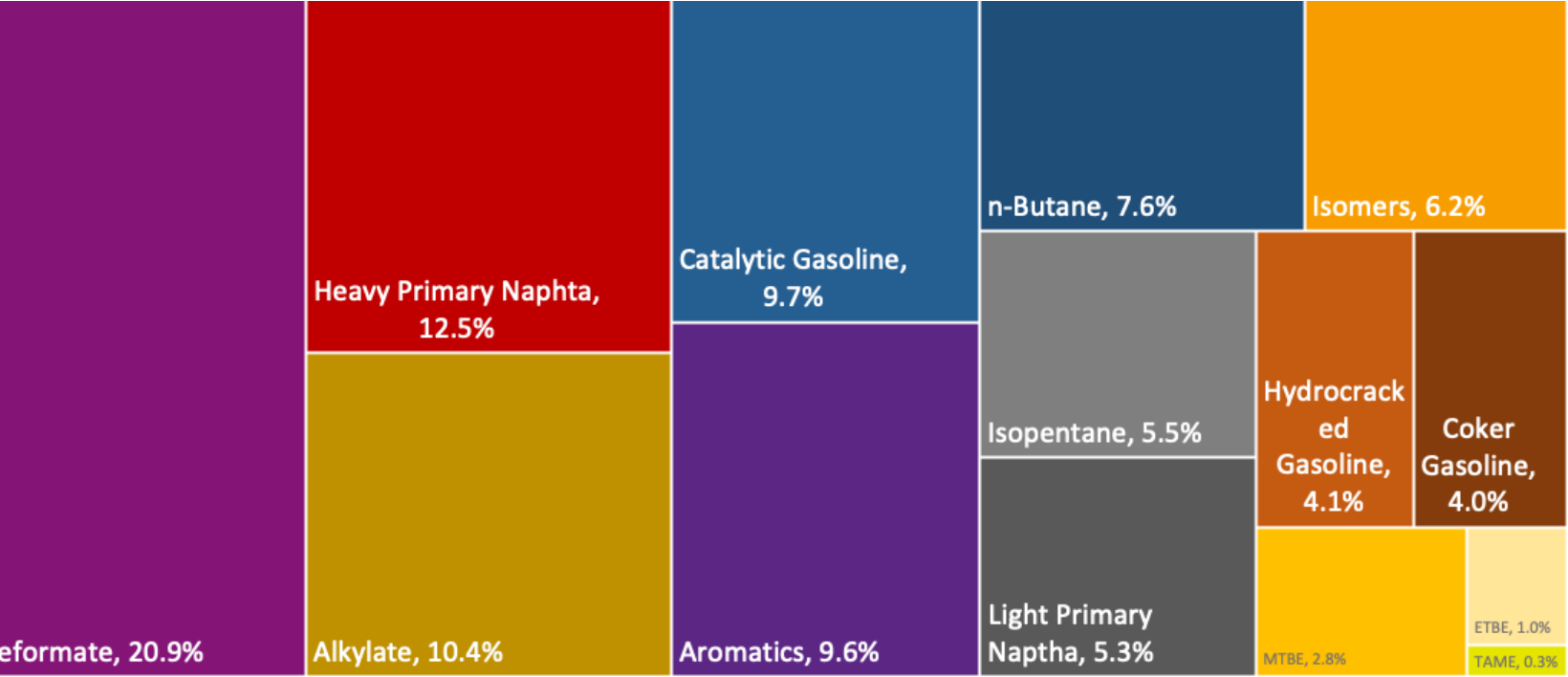
Gasoline Quality in Finland

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	16.5							
Advanced Biofuels (%cal)	4.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

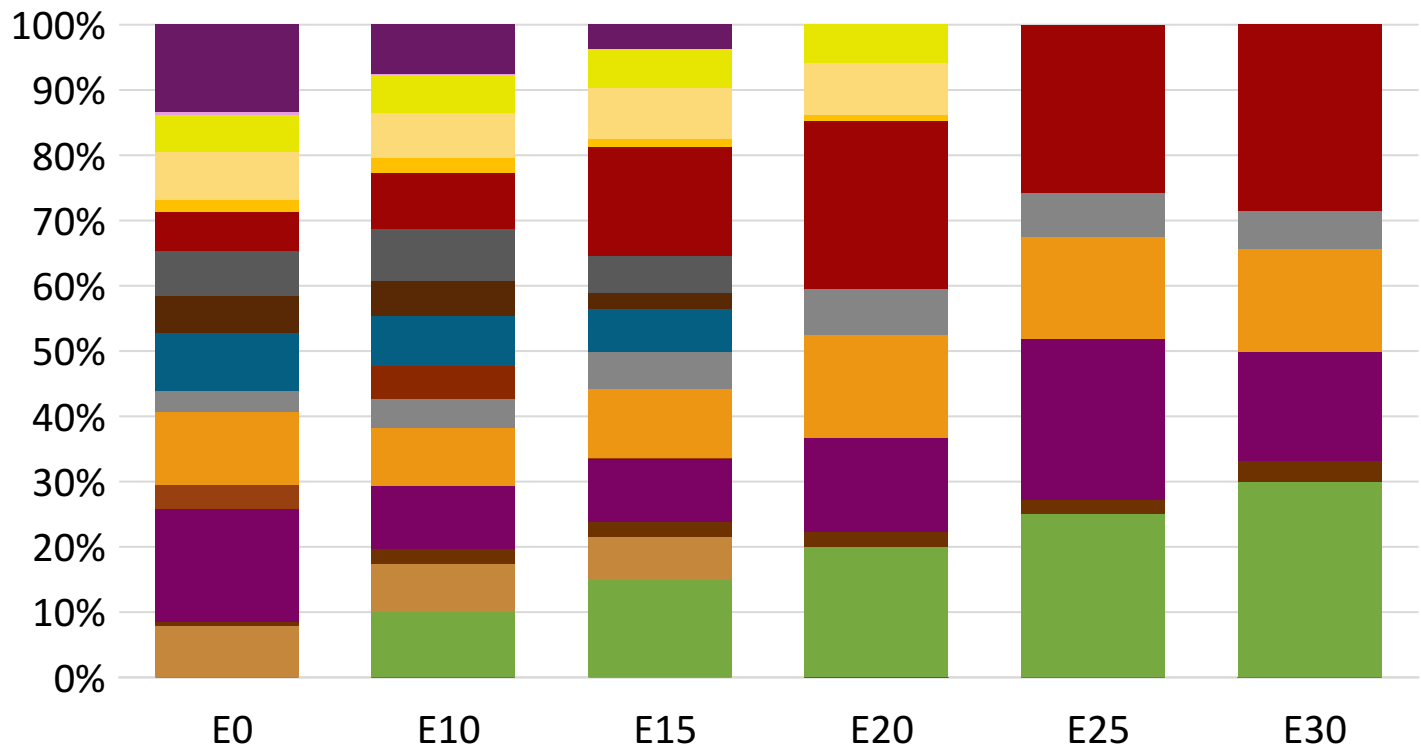
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Finland Available Blending Components



Finland – Gasoline 95 – Constant Octane

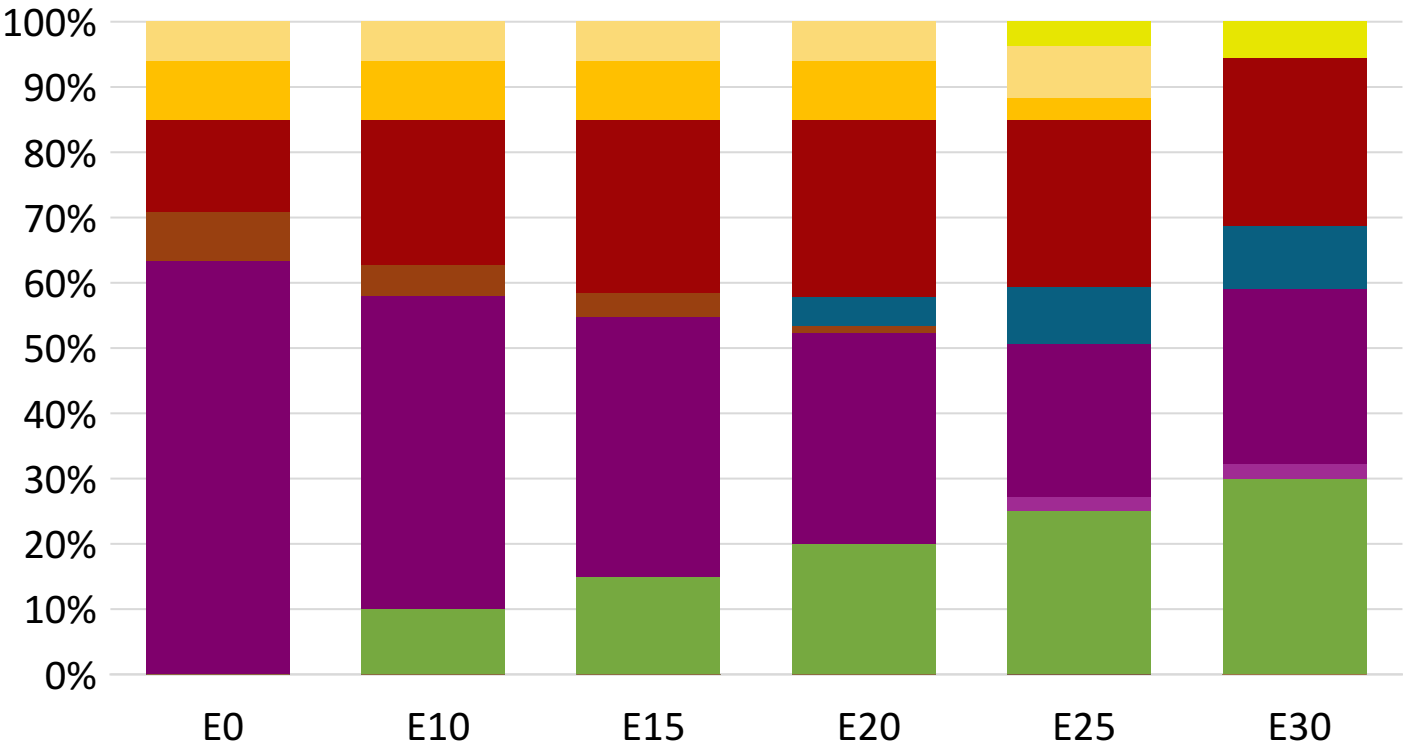
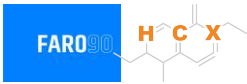


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.294	\$ 2.156	\$ 2.056	\$ 1.941	\$ 1.895	\$ 1.842

Source: Faro90

Finland - Gasoline 98 - Constant Octane

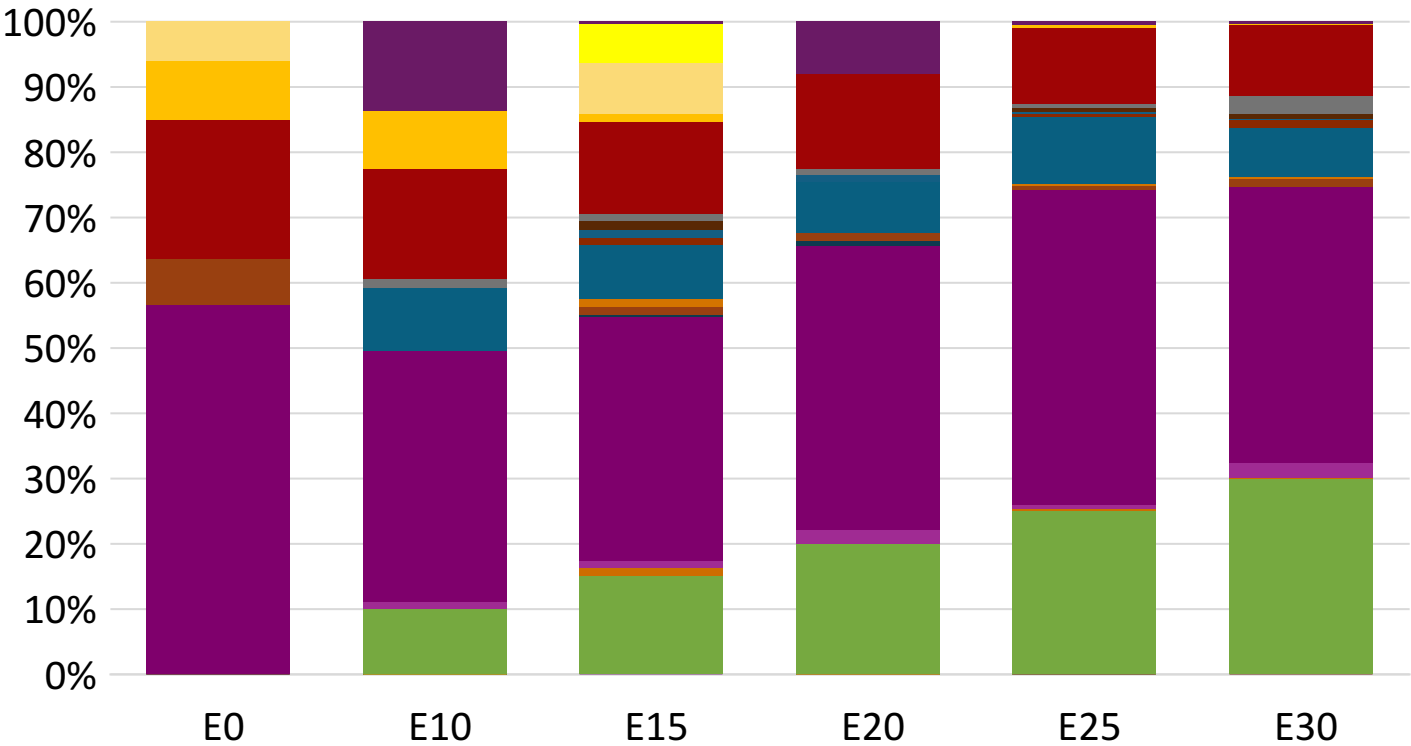
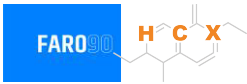


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.314	\$ 2.197	\$ 2.128	\$ 2.056	\$ 1.991	\$ 1.941

Source: Faro90

Finland – Gasoline 95 – Increased Octane

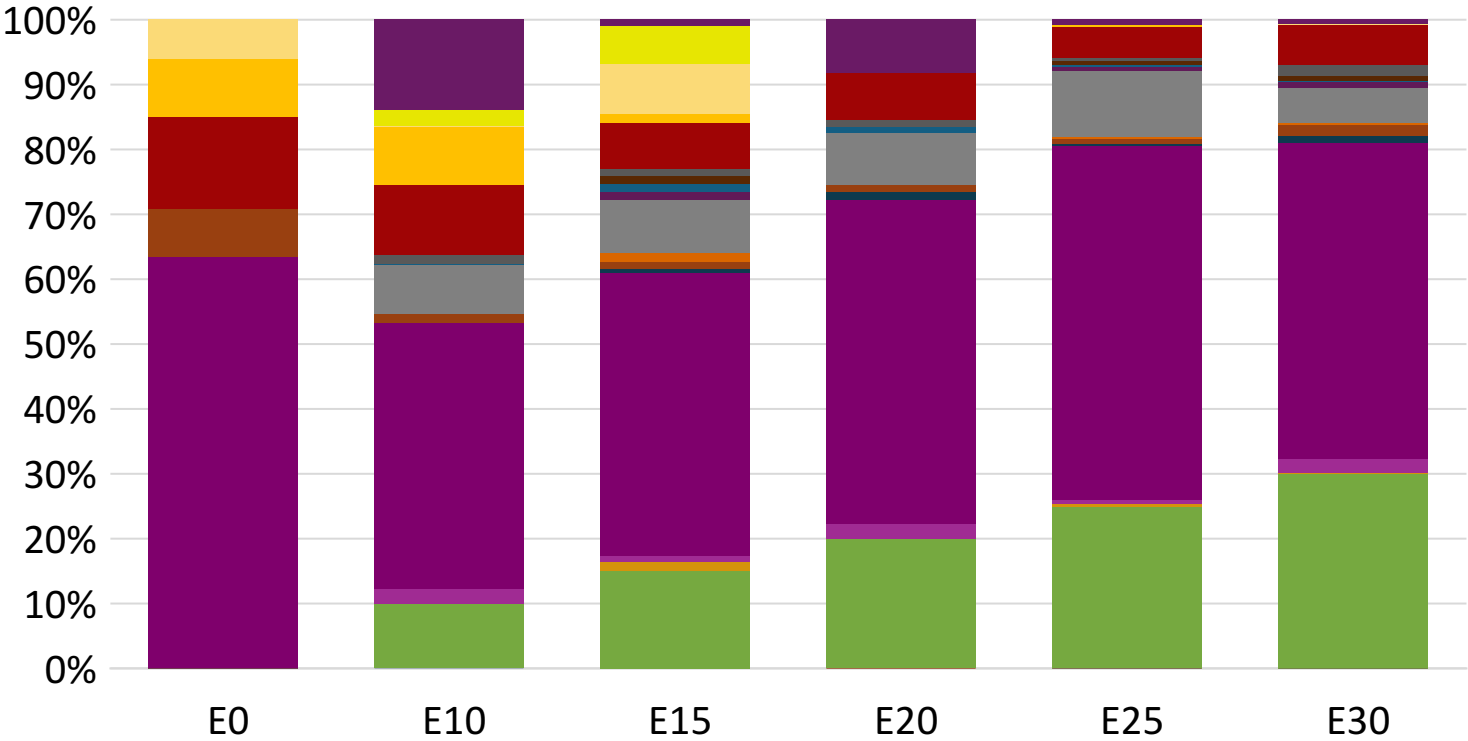


Average CIF 2024 Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.182	\$ 2.182	\$ 2.182	\$ 2.182	\$ 2.182	\$ 2.182

Source: Faro90

Finland – Gasoline 98 – Increased Octane

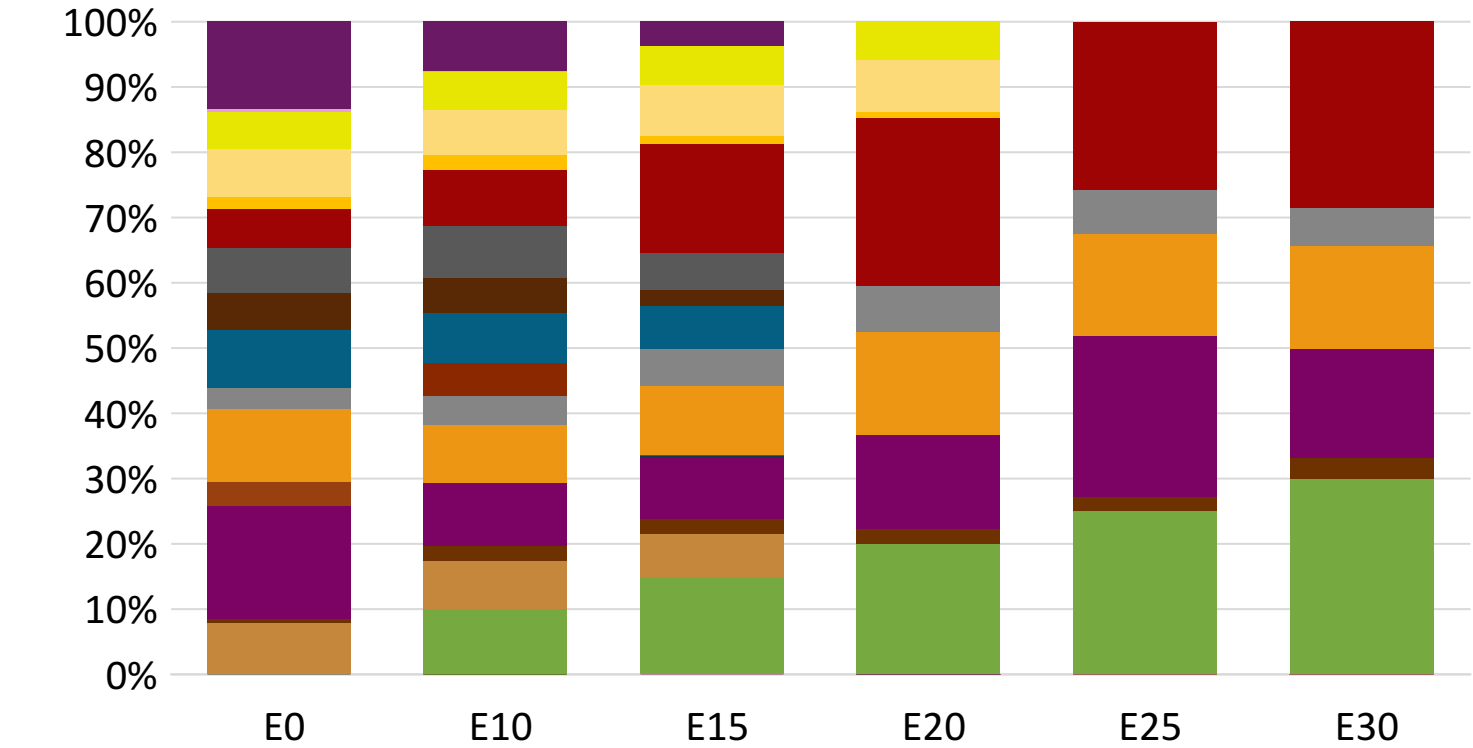


Average CIF 2024 Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.314	\$ 2.314	\$ 2.314	\$ 2.314	\$ 2.314	\$ 2.314

Source: Faro90

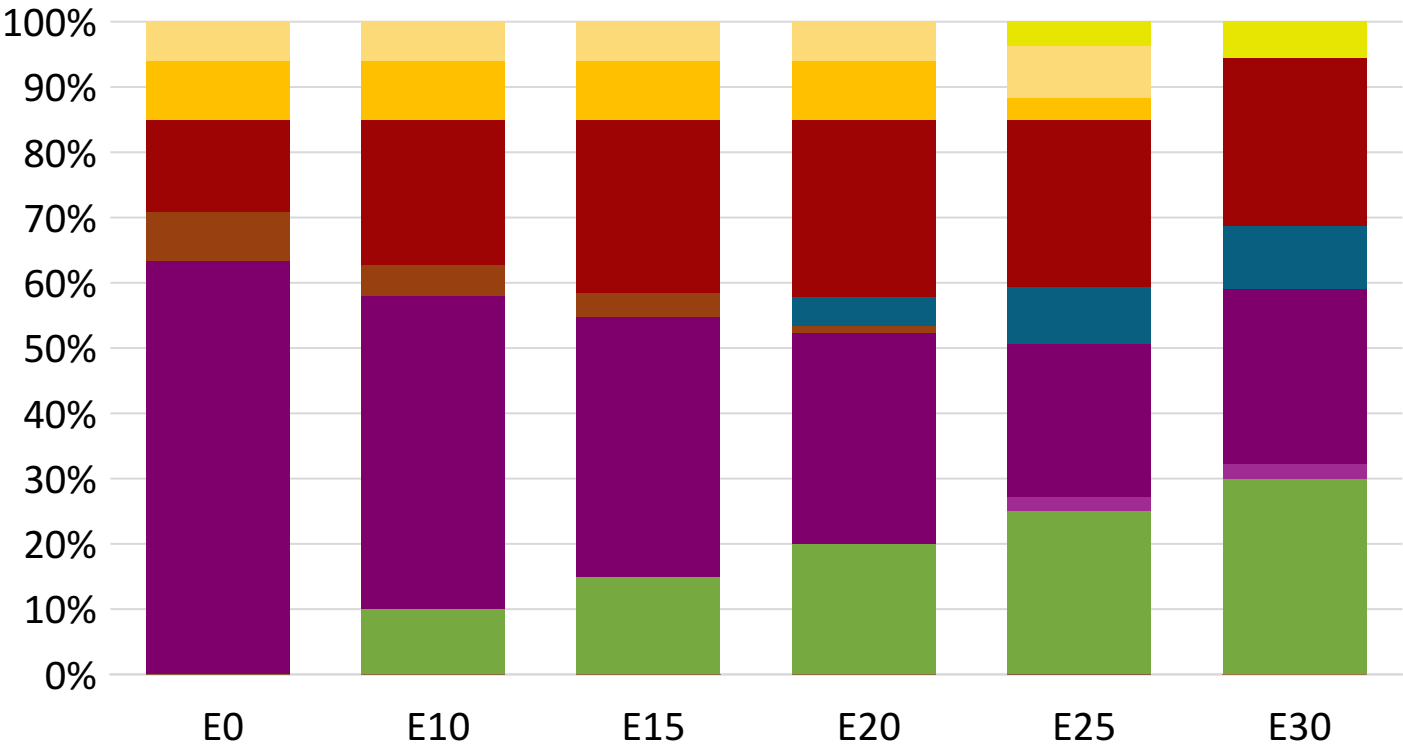
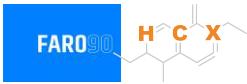
Finland – Gasoline 95 – Constant Octane



Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.274	\$ 2.136	\$ 2.036	\$ 1.921	\$ 1.875	\$ 1.822

Finland - Gasoline 98 - Constant Octane



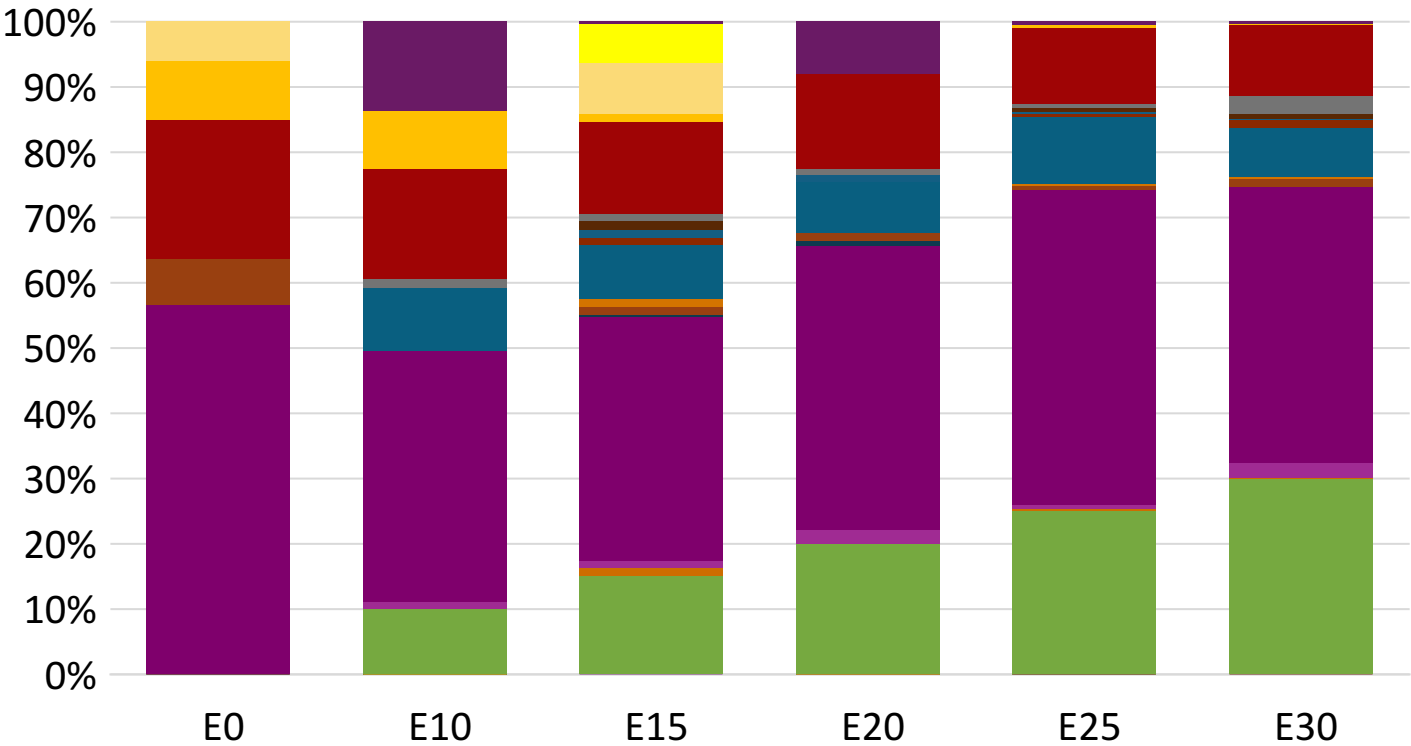
- Ethanol
- Alkylate
- Raffinate
- Reformate
- n-Butane
- Isomerate
- Isopentane
- Hydrocracked Gasoline
- Catalytic Gasoline
- Coker Naphtha
- Light Primary Naphtha
- Heavy Primary Naphtha
- MTBE
- ETBE
- TAME
- Aromatics

Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.294	\$ 2.177	\$ 2.108	\$ 2.036	\$ 1.971	\$ 1.921

Source: Faro90

Finland – Gasoline 95 – Increased Octane

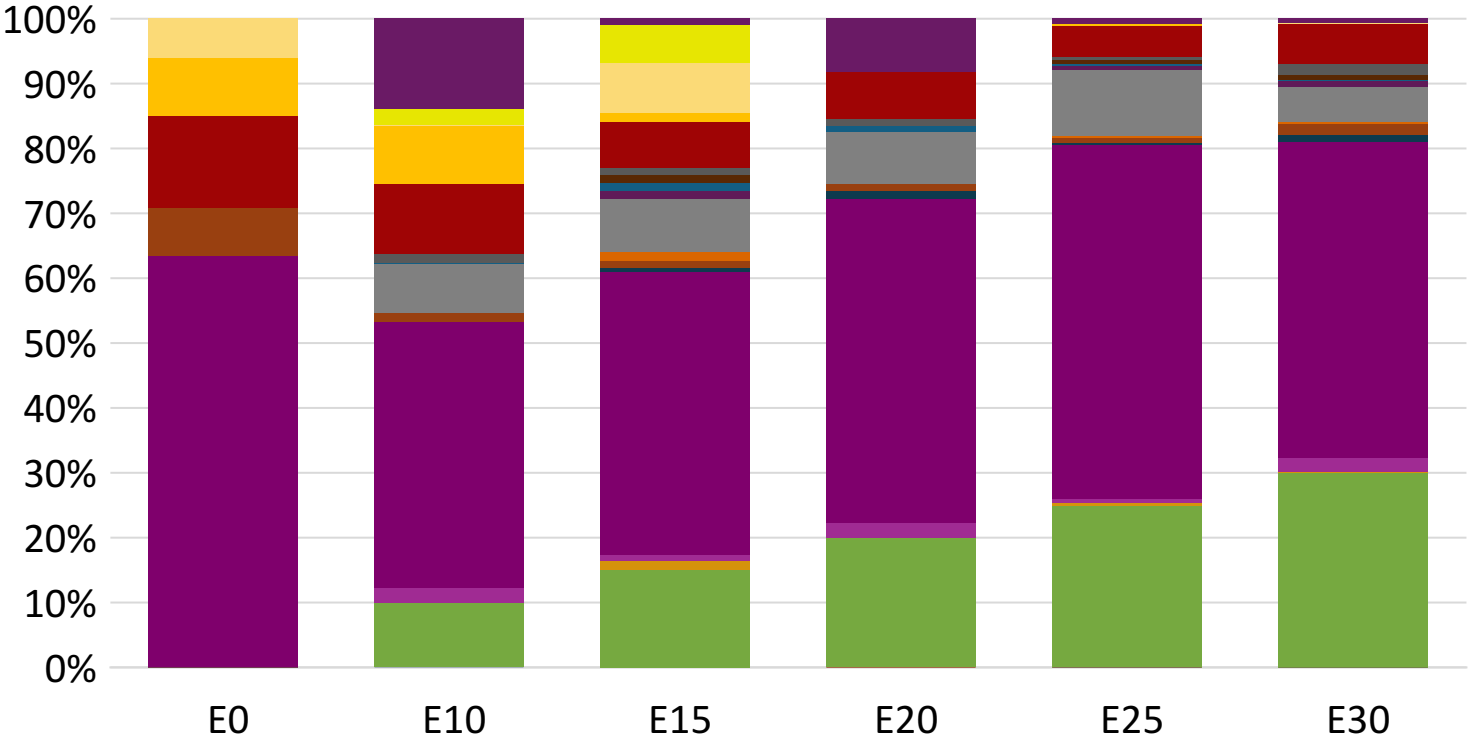


Average CIF 2024 Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162

Source: Faro90

Finland – Gasoline 98 – Increased Octane



Average CIF 2024 Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

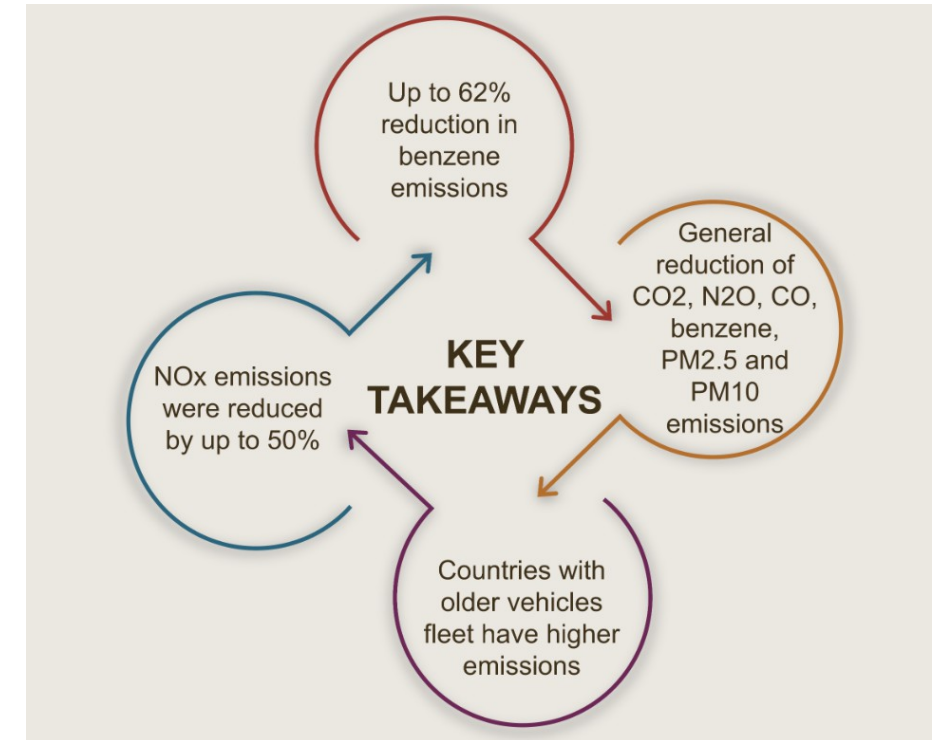
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

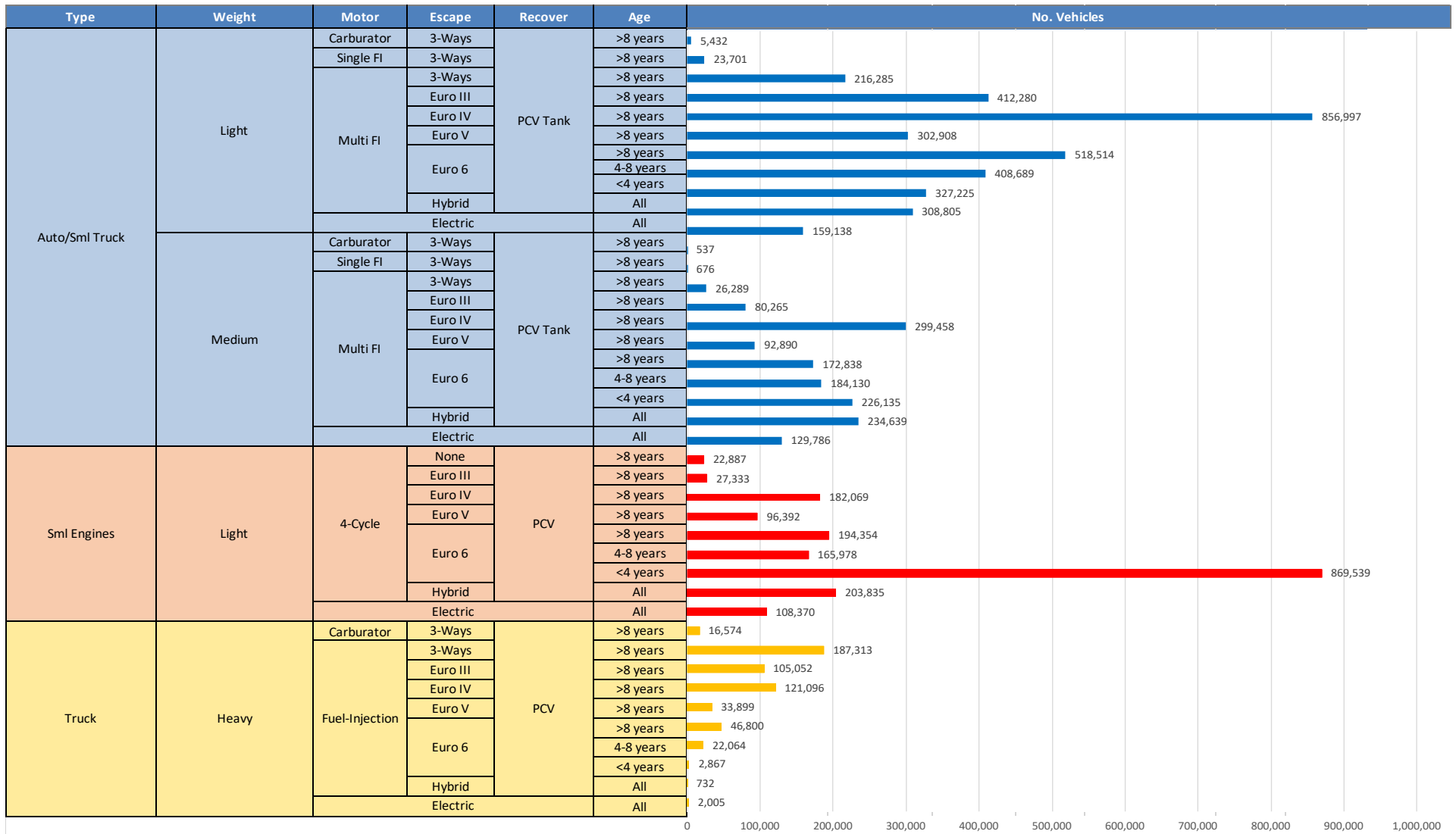
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

Gasoline Vehicle Fleet - Finland

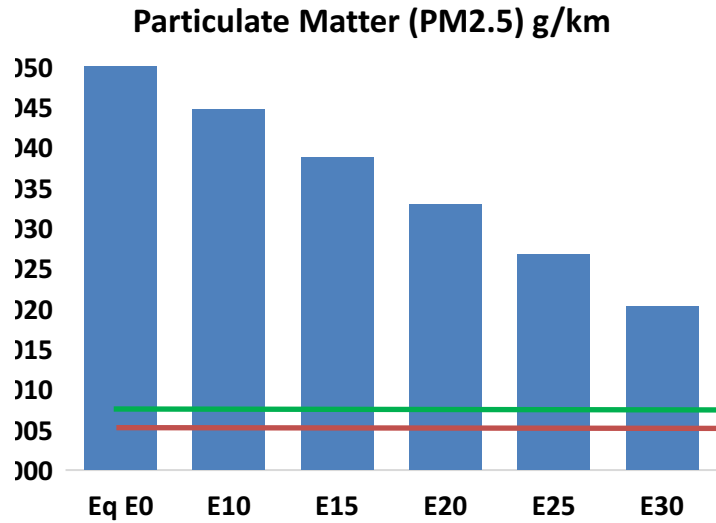
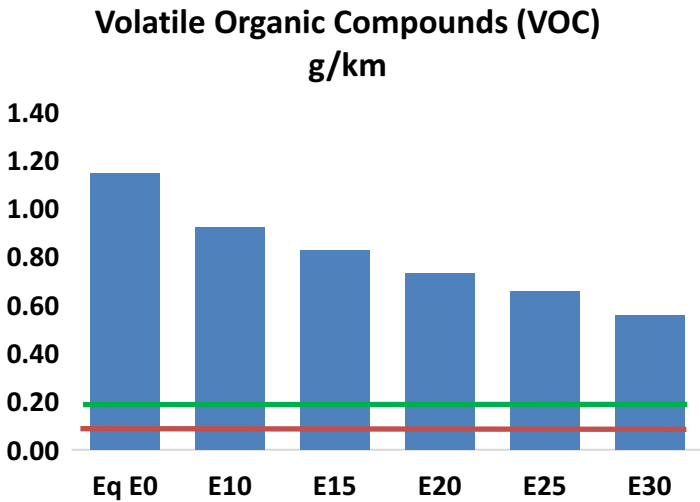
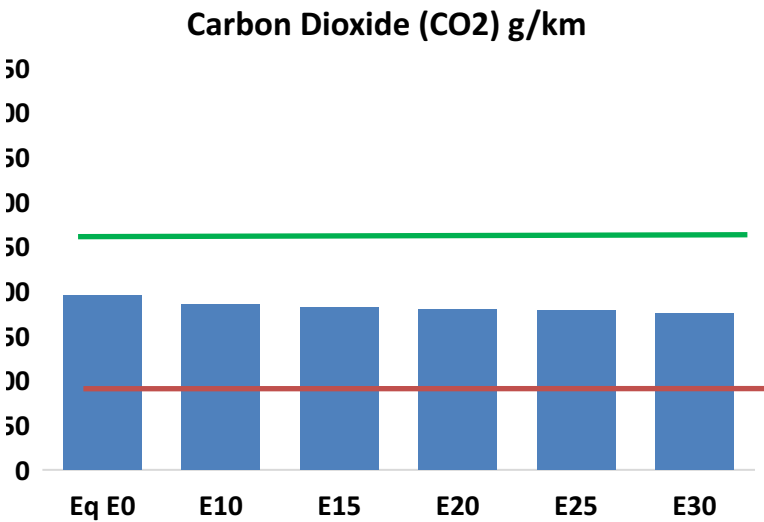
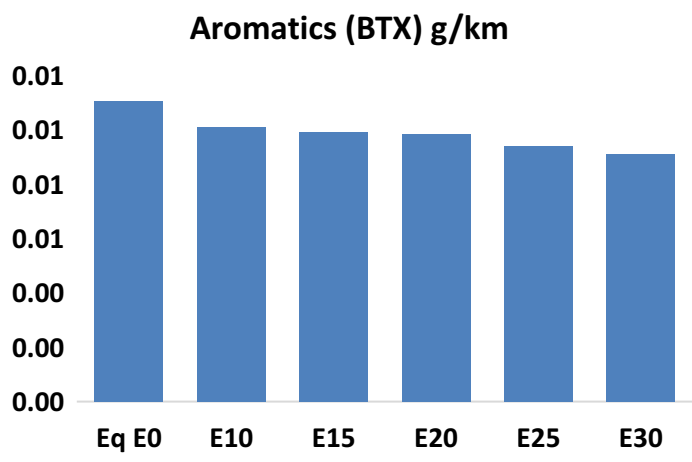
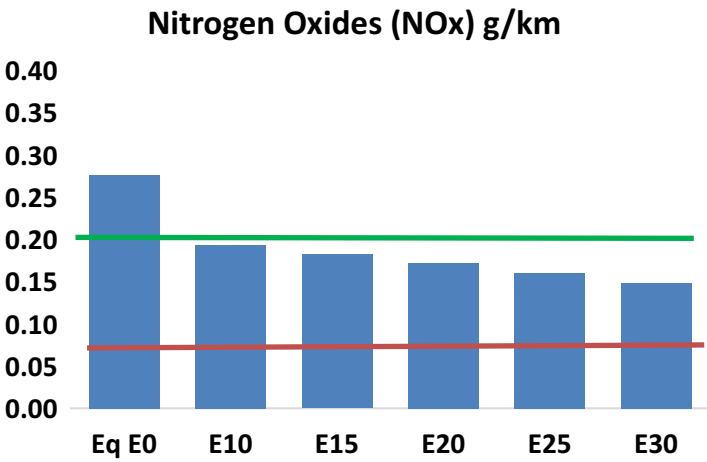
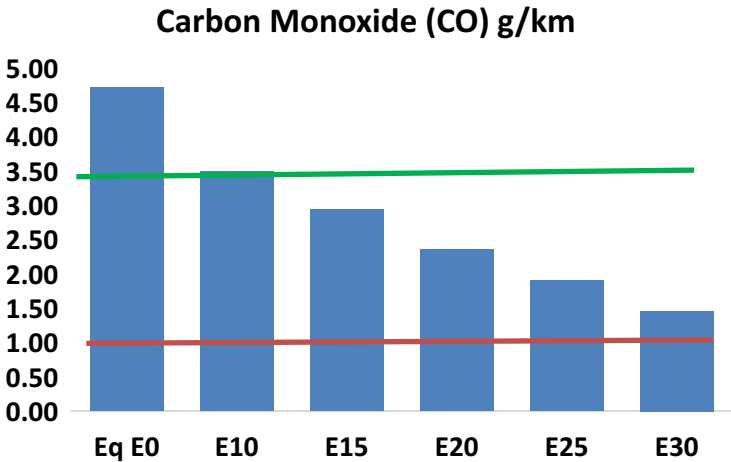
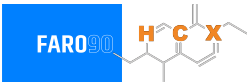


Vehicle Fleet: 7,396,775
Gasoline Vehicle Fleet: 3,837,401

Source: Eurostat, ACEA

Finland – Vehicular Emissions

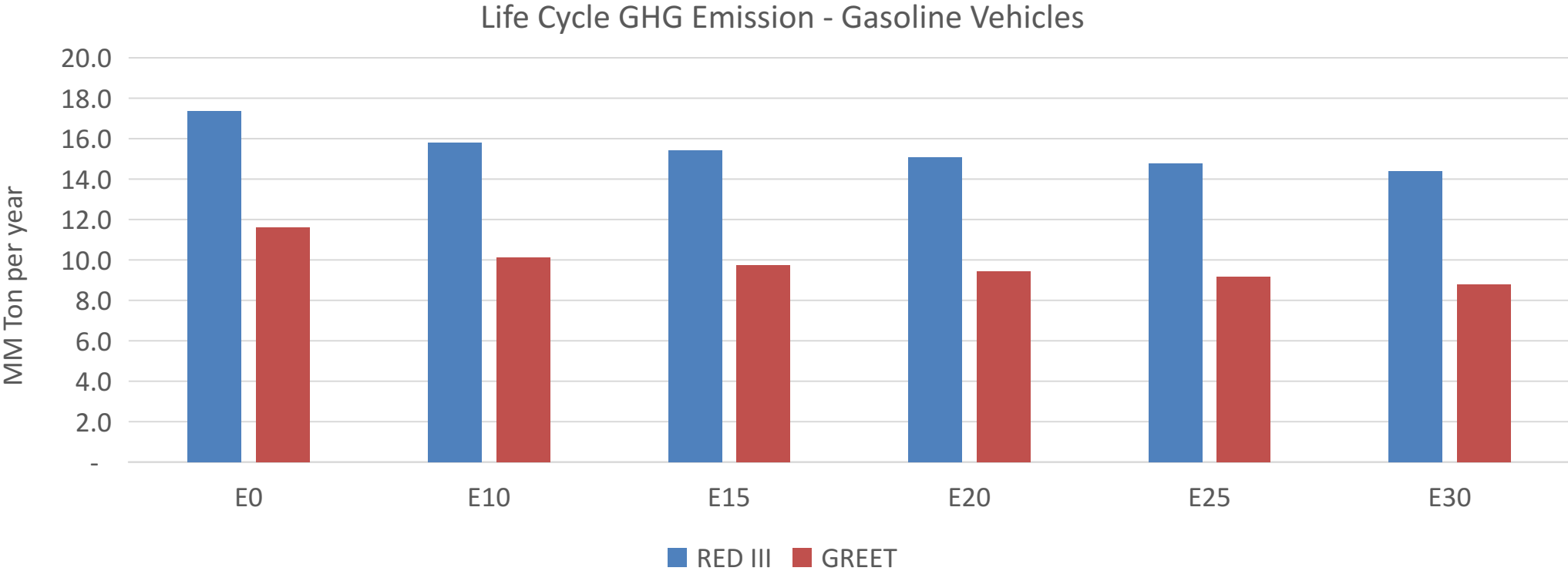
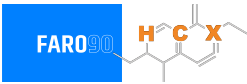
TIER III BIN 125 United States
Euro 6 Standard



Finland – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	469,869	408,719	347,570	292,577	234,989	190,028	145,320	-26%	-50%
CO2	19,452,404	18,952,199	18,479,784	18,108,243	17,924,390	17,818,561	17,490,113	-5%	-8%
VOC	114,075	103,064	92,053	82,593	72,871	65,335	55,493	-19%	-36%
NOx	27,517	20,638	19,262	18,154	17,121	15,982	14,777	-30%	-38%
SOx	221	204	188	174	160	145	130	-15%	-28%
NH3	7,866	7,788	7,710	7,747	7,834	7,895	7,946	-2%	0%
N2O	324	322	321	323	329	333	337	-1%	2%
CH4	22,977	22,977	22,977	23,322	23,828	24,264	24,678	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	467	434	400	373	347	326	299	-14%	-26%
Acetaldehyde	1,791	2,056	3,017	4,594	6,259	7,152	8,451	68%	249%
Formaldehyde	7,151	6,952	8,104	9,516	9,969	11,029	12,006	13%	39%
Benzene	1,102	1,059	1,003	987	979	936	906	-9%	-11%
PM2.5	1,230	1,092	954	828	703	571	433	-22%	-43%
PM10	4,504	3,999	3,494	3,035	2,577	2,091	1,585	-22%	-43%

Finland – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	64.9
Target @ 2035 (MMT/year)	36.6
Delta	28.4

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.7%	15.9%	18.5%	21.0%	24.2%
RED II/III	0.0%	8.9%	11.3%	13.1%	14.9%	17.2%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.2%	6.5%	7.6%	8.6%	9.9%
RED II/III	0.0%	5.5%	6.9%	8.0%	9.1%	10.5%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90